gwflow

Purpose

The gwflow module simulates groundwater on a grid of cells and replaces the one-dimensional aquifers. It runs when `gw_flow = 1` in codes.bsn and writes the output files described here. The input files are described on the gwflow input page.

The files fall into these groups:

- basin and cell water balance (`gwflow_basin_wb_*`, `gwflow_cell_wb_*`)
- heat and solute balances (`gwflow_basin_heat_*`, `gwflow_basin_sol_*`)
- observation wells (`gwflow_obs_*`)
- HRU pumping (`gwflow_hru_pump_*`)
- diagnostic files for canals, ponds, tiles and channels
- channel hydrograph separation (`gwflow_chan_hydsep_day.txt`)
- the cell definition and the run log, which are always written

Output switches

The `write_day`, `write_mon`, `write_yr` and `write_aa` flags in codes.gw (1 on, 0 off) switch the water balance files. The heat and solute balance files follow the same flags, and the hydrograph separation file follows `write_day`. Observation well, pumping and diagnostic output is on by default.

print.prt can override these settings with six rows, each with the usual daily, monthly, yearly and average annual flags. The rows are recognised only when print.prt uses object labels (`use_obj_lbls = y`). A row that is present replaces the default; a row that is absent leaves it as it is. All overrides are applied before any gwflow output file is opened, so a file that is switched off is not created.

print.prt row	Effect
`gwflow_wb`	each flag switches the water balance file for that interval; heat, solute and hydrograph separation files follow it unless their rows are present
`gwflow_heat`	each flag switches the heat balance file for that interval (there is no monthly heat file)
`gwflow_solute`	each flag switches the solute balance files for that interval
`gwflow_obs`	each flag switches the observation well file for that interval
`gwflow_pump`	any flag set to `y` switches on all HRU pumping files
`gwflow_flux`	the daily or yearly flag switches on the canal, pond, tile and channel diagnostic files

Example rows that keep only the monthly water balance and the daily observation wells:

```
objects                daily       monthly        yearly         avann
gwflow_wb                  n             y             n             n
gwflow_obs                 y             n             n             n
```

gwflow_pump	n	n	n	n
gwflow_flux	n	n	n	n

Basin water balance

gwflow_basin_wb_day.txt, _mon.txt, _yr.txt and _aa.txt give the water balance of the whole grid. Volumes are summed over all active cells and divided by the watershed area, so they are in mm over the watershed. Line 1 is a title, line 2 the column header and line 3 the units. The seven leading columns are jday, mon, day, yr, unit, gis_id and name (the project name).

The daily file has these columns after the leading ones:

Column	Units	Description
ts	day	gwflow time step
vbef	mm	groundwater volume at the start of the day
vaft	mm	groundwater volume at the end of the day
recharge	mm	recharge from the soil profile
gw_et	mm	groundwater evapotranspiration
gw_sw	mm	groundwater discharge to channels
sw_gw	mm	channel seepage to groundwater
sat_excess	mm	saturation excess flow to the land surface
soil	mm	groundwater moved into the soil profile
lateral	mm	net lateral flow between cells
boundary	mm	exchange across the grid boundary
pump_allo	mm	pumping for water allocation (irrigation)
pump_ext	mm	pumping specified in pumpex.gw
tile	mm	tile drain outflow
reservoir	mm	exchange with reservoirs
wetland	mm	exchange with wetlands
canal	mm	canal seepage
floodplain	mm	exchange with floodplains
pond	mm	recharge pond seepage
phytorem	mm	phreatophyte transpiration
error	%	volume balance error
sat_frac	fraction	fraction of active cells with saturation excess flow that day
wt_depth	m	average depth to the water table (see the note below)
pump_def	mm	allocation pumping demand that could not be met

The monthly, yearly and average annual files replace ts, vbef and vaft with dvol (change in storage, mm) and drop error, sat_frac and wt_depth. They end with pump_def after phytorem.

The daily wt_depth is the average over the active cells. In the average annual file, dvol is the total change in storage over the run divided by the number of years, and every flux is the run total divided by the number of years.

Cell water balance

gwflow_cell_wb_day.txt, _mon.txt, _yr.txt and _aa.txt have one row per active cell per interval. Lines 1 and 2 are notes on units; line 3 is the column header.

Column	Units	Description
jday, mon, day, yr	-	date of the row
unit	-	cell number in the gwflow grid
gis_id	-	cell ID from the gis_id column of cells.gw (the gwflow cell shapefile ID)
name	-	cell name from cells.gw, such as cell0001 (first 12 characters)
head	m	groundwater head
wt_depth	m	land surface elevation of the cell minus the head
recharge	m^3/day	recharge

Column	Units	Description
gw_et	m³/day	groundwater evapotranspiration
gw_sw	m³/day	discharge to channels
sw_gw	m³/day	channel seepage
sat_excess	m³/day	saturation excess flow
soil	m³/day	transfer to the soil profile
lateral	m³/day	net lateral flow
pump_allo	m³/day	allocation pumping
pump_ext	m³/day	specified pumping
tile	m³/day	tile drain outflow
reservoir	m³/day	reservoir exchange
wetland	m³/day	wetland exchange
floodplain	m³/day	floodplain exchange
canal	m³/day	canal seepage
pond	m³/day	recharge pond seepage
phytorem	m³/day	phreatophyte transpiration

The monthly, yearly and average annual files give the average daily rate over the interval, so all fluxes stay in m³/day.

wt_depth is measured from the cell's land surface elevation, which on a coarse grid represents the valley and not the streambed. Cells along channels can therefore show a deep water table while the head is close to the channel level. This comes from the grid resolution; the head itself is not affected.

Heat balance

When heat transport is on (heat = 1 in codes.gw), gwflow_basin_heat_day.txt, _yr.txt and _aa.txt are written. They follow the daily, yearly and average annual water balance switches, or the gwflow_heat row of print.prt when it is present; there is no monthly heat file. Values are in MJ.

The daily file has ts, hbef and haft (heat stored at the start and end of the day), then recharge, gw_et, gw_sw, sw_gw, sat_excess, soil, lateral, dispersion, boundary, pump_allo, pump_ext, tile, reservoir, wetland, canal, floodplain, pond, then error (%) and tavg (average groundwater temperature, °C). The yearly and average annual files start with hdel (change in heat stored) and end at pond. ## Solute balances {#output-reference-gwflow-solute-balances}

When solute transport is on (solute = 1 in codes.gw), one file per solute and interval is written, named gwflow_basin_sol_<solute>_<interval>.txt. Nitrate (no3) and phosphorus (p) are always included; with salts in the model, so4, ca, mg, na, k, cl, co3 and hco3 are added; with constituents, seo4 and seo3. Values are in kg. The interval files follow the water balance switches, or the gwflow_solute row of print.prt when it is present.

The daily file has ts, mbef and maft (solute mass at the start and end of the day), then recharge, gw_sw, sw_gw, sat_excess, soil, advection, dispersion, react_in, react_out, mineral, sorption, pump_allo, pump_ext, tile, reservoir, wetland, canal, floodplain, pond and error (%). The monthly, yearly and average annual files start with delm (change in mass) and end at pond.

Observation wells

Observation cells are listed as observation_cell lines in outputs.gw. The files gwflow_obs_day.txt, _mon.txt, _yr.txt and _aa.txt have one row per observation cell:

Column	Units	Description
unit	-	observation well number, in the order listed in outputs.gw
gis_id	-	cell number of the observation cell
name	-	obs_0001, obs_0002, …
head	m	groundwater head
wt_depth	m	land surface elevation minus head
temperature	°C	groundwater temperature; -99 when heat transport is off
no3	mg/L	nitrate concentration; -99 when solute transport is off

Column	Units	Description
p	mg/L	phosphorus concentration; -99 when solute transport is off

The monthly and yearly files give averages over the interval, and the average annual file the average over the run.

HRU pumping

When pumping for water allocation is simulated, **gwflow_hru_pump_day.txt**, **_mon.txt**, **_yr.txt** and **_aa.txt** report the groundwater pumped for each HRU in the **pump_allo** column. The daily file is in m^3/day and lists only HRUs that pumped that day; the monthly and yearly files give totals in m^3, and the average annual file m^3/yr.

When hru_pump.gw lists HRUs to observe, **gwflow_cell_wb_ppag_obs_day.txt** gives their daily pumping in m^3, one column per listed HRU. It is written under the same switch as the other pumping files.

Diagnostic files

These files are written when the related feature is in use and diagnostic output is on (**gwflow_flux**):

File	Content
gwflow_canal_wb_day.txt	daily canal water balance in m^3: **diversion**, **storage**, **pond_xfer**, **seepage**
gwflow_canal_sol_day.txt	daily canal solute balance in kg: **div_mass**, **storage**, **pond_mass**, **seep_mass** for each solute
gwflow_pond_wb_day.txt	daily recharge pond water balance: **area** (m^2), **storage**, **rain**, **div_added**, **evap**, **recharge**, **div_spec**, **div_unsat** (m^3)
gwflow_pond_sol_day.txt	daily pond solute balance: **area**, **storage**, **solute**, **mass**, **div_added**, **recharge** (kg)
gwflow_pond_mass_day.txt, gwflow_pond_conc_day.txt	daily solute mass (kg) and concentration (g/m^3), one column per pond
gwflow_tile_group_day.txt	daily tile flow (m^3/s) for each group of tile cells
gwflow_gwsw_group_day.txt	daily groundwater and channel exchange (m^3) for each channel cell group in sw_group.gw
gwflow_chan_obs_flow_day.txt	daily groundwater and channel exchange (m^3) at the channel cells flagged in chancell.gw
gwflow_chan_obs_no3_day.txt	the same for nitrate (g), when solute transport is on

When **gwflow.wbgroups** is present, one daily water balance file per cell group is also written, named **gwflow_group_wb_day_** followed by the group number as two characters (a space and the digit for groups 1 to 9). Its columns follow the daily basin file, in m^3.

Hydrograph separation

gwflow_chan_hydsep_day.txt is written when daily water balance output is on. It gives, for each channel and day, the flow from each source in m^3/s: **surf_runoff**, **lateral**, **gw_discharge**, **stream_seep**, **satex_gw**, **satex_sw** and **tile_drain**.

Files always written

File	Content
gwflow_record	run log: which input files were found, which features are active, which print.prt rows changed the output switches, and error messages from reading
gwflow_cell_definition.txt	one line per cell: **cell_id**, **row**, **col**, **x_coord**, **y_coord**, **zone**, **status**, **area**

When transit.gw is present on a structured grid, **gwflow_transit_cell**, **gwflow_transit_chan** and (with tiles) **gwflow_transit_tile** give travel times in days.

The **head_output_time** lines in outputs.gw are still read but no longer produce output; cell heads are in the cell water balance files.

Related files

- gwflow input files
- codes.bsn: gw_flow switch
- print.prt: output rows
- Aquifer theory

Sources

- Output file setup and basin, cell, heat, solute and observation writers: gwflow_output.f90 (gwflow_output_init, gwflow_output_day, the monthly, yearly and average annual routines, gwflow_write_celldef)
- codes.gw flags, print.prt overrides, pumping, diagnostic and hydrograph separation file setup: gwflow_read.f90
- Flags and unit numbers: gwflow_module.f90
- Daily simulation, pumping output and the call to the average annual output: gwflow_simulate.f90; pond and canal diagnostics in gwflow_pond.f90 and gwflow_canal_div.f90
- print.prt rows: basin_print_codes_read.f90; gwflow_record opened in basin_read_objs.f90

Switch	Fortran variable
write_day, write_mon, write_yr, write_aa in codes.gw, or the gwflow_wb row	gwflag_day, gwflag_mon, gwflag_yr, gwflag_aa
gwflow_heat row (default: water balance flags)	gwflag_heat_day, gwflag_heat_yr, gwflag_heat_aa
gwflow_solute row (default: water balance flags)	gwflag_solute_day, gwflag_solute_mon, gwflag_solute_yr, gwflag_solute_aa
gwflow_obs row (default: all on)	gwflag_obs_day, gwflag_obs_mon, gwflag_obs_yr, gwflag_obs_aa, and gwflag_obs as their maximum
gwflow_pump, gwflow_flux rows (default: on)	gwflag_pump, gwflag_flux
hydrograph separation	hydsep_flag, set from gwflag_day
heat, solute in codes.gw	gw_heat_flag, gw_solute_flag

Cell file column	Fortran variable
gis_id, name	cell_gis_id(i), cell_name(i) (written with a12)
recharge … phytorem	gw_hyd_ss(i)%rech, gwet, gwsw, swgw, satx, soil, latl, ppag, ppex, tile, resv, wetl, fpln, canl, pond, phyt